

The Basin Depth–Sleep–Longevity Theorem

Sleep Requirement as a Readout of Attractor Landscape Position: A Unified Geometric Framework Spanning Cellular Biology to 500 Million Years of Evolutionary Evidence

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Abstract

Sleep duration varies by orders of magnitude across the animal kingdom, from zero in the horseshoe crab (*Limulus polyphemus*, lineage ~ 450 Mya) and functionally zero in the Greenland shark (*Somniosus microcephalus*, lifespan up to 400 years, the longest-lived vertebrate) to approximately eight hours of REM-rich sleep in humans. The prevailing explanation treats sleep as a universal biological necessity with conserved functions. We propose and defend an alternative: **sleep is a cost-discharge mechanism, not a fundamental necessity**. Sleep requirement is proportional to the sum of three cost axes that accumulate during waking — oxidative/metabolic cost (Axis 2), attractor switching cost (Axis 1), and synaptic/computational cost (Axis 3) — and approaches zero when all three approach zero. This occurs when an organism occupies a deep, stable attractor basin (the Lock strategy). We formalise this as the *Basin Depth–Sleep–Longevity Theorem*, which makes five falsifiable predictions confirmed by the comparative and evolutionary record. The decisive experimental proof is provided by *Astyanax mexicanus* cavefish (North et al., 2024), in which sleep was lost convergently and independently in at least three cave populations from a surface-sleeping ancestor upon colonisation of a deep, stable, no-light niche. The theorem unifies sleep biology, longevity biology, and evolutionary theory within the Waddington attractor landscape framework ($S + N + G \rightarrow R$), and identifies a triad of geometric variables — basin depth, sleep requirement, and lifespan — as co-expressions of the same underlying thermodynamic quantity: position-dependent cost accumulation rate in a biological landscape.

Keywords: sleep geometry; attractor landscape; Lock strategy; longevity; Waddington; Greenland shark; cavefish; ROS; oxidative cost; evolutionary convergence; basin depth;

S+N+G

Repository: <https://github.com/Eric-Robert-Lawson/attractor-oncology>

Preamble: The Central Claim

Sleep is not a fundamental biological necessity.

Sleep is a cost-discharge mechanism. The necessity is the cost. Eliminate the cost and you eliminate the requirement for the solution.

This claim is falsifiable. It is stated formally in Section 2 and tested against the full comparative record in Section 4.

1 Theoretical Background: The Attractor Framework

1.1 The Waddington Landscape and $S + N + G = R$

All biological systems — cells, organisms, and species — exist within an epigenetic attractor landscape formalised by Waddington [1]. In the present framework (Lawson, OrganismCore, 2025–2026), the landscape is characterised by three quantities:

- **S**: the intrinsic state of the system (genome, epigenome, chromatin accessibility)
- **N**: the external field acting on the system (developmental signals, social environment, circadian light-dark cycle, oncogenic signals)
- **G**: the resultant landscape geometry (the topology of attractor basins available to the system given S and N)

The product is **R**: the observable state — which basin the system occupies and how deep within that basin it sits.

This framework was developed for cancer (the Identity Axis Theorem, IAT; Lawson 2026a), validated in 36 cancer types with zero biological contradictions, five FDA drug confirmations, and fourteen Phase 1–3 trial confirmations. It was subsequently derived independently for sleep (Lawson 2026b), domestication (Lawson 2025a, 2025b), and neurodegenerative disease (Lawson 2025c).

The present paper applies the framework to the full cross-species sleep record, integrating it with longevity data and deep-time evolutionary evidence.

1.2 The Lock and Explore Strategies

Two archetypal landscape positions exist:

Definition 1 (Lock Strategy). *An organism occupying a deep, stable attractor basin with minimal switching cost, minimal metabolic rate, and minimal computational navigational load. Characteristic of ancient lineages in stable ecological niches.*

Definition 2 (Explore Strategy). *An organism occupying a shallow, variable attractor basin with high switching cost, high metabolic rate, and high computational navigational load. Characteristic of recent lineages in complex, variable ecological niches.*

These are poles of a continuous spectrum, not discrete categories.

2 The Theorem

2.1 The Three Cost Axes

Sleep discharges costs that accumulate during waking. Three primary cost axes have been identified:

Definition 3 (Axis 1: Switching Cost, C_1). *The epigenetic and metabolic cost of transitioning between attractor states: threat response, decision-making, social navigation, environmental repositioning. High in organisms with shallow, variable landscapes. Near-zero in organisms with deep, stable landscapes.*

Definition 4 (Axis 2: Oxidative/Metabolic Cost, C_2). *Reactive oxygen species (ROS) accumulation during aerobic metabolism, primarily in gut epithelium and neurons. Proportional to metabolic rate and body temperature. Near-zero in cold-water, low-metabolic-rate organisms.*

Definition 5 (Axis 3: Synaptic/Computational Cost, C_3). *Synaptic weight accumulation requiring offline consolidation and downscaling that cannot occur during active navigation. Proportional to social landscape complexity, learning demand, and spatial navigational load. Near-zero in organisms with simple behavioural programmes.*

2.2 The Total Sleep Requirement Function

$$S_{\text{total}} = f(C_1 + C_2 + C_3) \tag{1}$$

where f is monotonically increasing and $S_{\text{total}} \rightarrow 0$ as $(C_1 + C_2 + C_3) \rightarrow 0$.

2.3 Formal Statement of the Theorem

Theorem 1 (Basin Depth–Sleep–Longevity Theorem). *Let $\mathcal{B}(O)$ denote the depth of the attractor basin occupied by organism O in its Waddington landscape. Let $S(O)$ be the total*

sleep requirement of O , and $L(O)$ be the maximum lifespan of O under natural conditions.

Then:

1. $S(O)$ is a decreasing function of $\mathcal{B}(O)$:

$$\frac{dS}{d\mathcal{B}} < 0 \quad (2)$$

2. $L(O)$ is an increasing function of $\mathcal{B}(O)$:

$$\frac{dL}{d\mathcal{B}} > 0 \quad (3)$$

3. $L(O)$ is a decreasing function of $S(O)$:

$$\frac{dL}{dS} < 0 \quad (4)$$

4. As $\mathcal{B}(O) \rightarrow \infty$, $S(O) \rightarrow 0$ and $L(O) \rightarrow L_{\max}$.

5. Sleep requirement is not encoded in the genome. It is encoded in the landscape. The same genome placed in a deeper basin will evolve reduced sleep requirement under selection.

2.4 The Oxidative Substrate: Physical Grounding

The proximate death mechanism from total sleep deprivation is ROS accumulation in the gut epithelium, not in the brain [2]. Administering gut-specific antioxidants rescues sleep-deprived *Drosophila* without restoring sleep, confirming that the cost — not the sleep itself — is what must be addressed.

This physically grounds the theorem: cold-water organisms with near-zero metabolic rates (Greenland shark, nautilus, coelacanth) do not accumulate gut ROS at a rate requiring a dedicated discharge window. The discharge problem never arises.

2.5 The Hard Floor

Even in the deepest-basin organisms, a cellular maintenance floor exists: every aerobic cell produces some ROS. This floor has been confirmed by the sponge *Ephydatia muelleri* [6], which has no nervous system yet exhibits nightly quiescence with sleep rebound — pure cellular maintenance, zero synaptic component.

The sleep requirement of any organism is therefore:

$$S_{\text{total}} = S_{\text{floor}} + f(C_1 + C_2 + C_3) \quad (5)$$

where $S_{\text{floor}} > 0$ is the irreducible cellular maintenance quiescence common to all aerobic multicellular life.

2.6 Falsification Criteria

The theorem is falsified by any of the following:

1. A deep-basin, low-metabolic-rate, low-complexity organism with a high sleep requirement.
2. A shallow-basin, high-metabolic-rate, high-complexity organism with near-zero sleep requirement.
3. A longest-lived species with a high metabolic rate and high sleep requirement.
4. A species that evolves *increased* sleep upon colonising a stable, low-variance niche.
5. Sleep rebound upon deprivation in the horseshoe crab, nautilus, or Greenland shark.

None of these has been observed.

3 Formal Predictions

Prediction 1. *Organisms in deep, stable, low-variance attractor basins will require minimal or no sleep.*

Prediction 2. *Sleep requirement will be inversely correlated with basin depth across the tree of life, and this relationship will be monotonic.*

Prediction 3. *When a surface-sleeping species colonises a deep-basin niche, it will evolve loss of sleep convergently — the same outcome will appear in every independent colonisation event.*

Prediction 4. *The oldest surviving lineages will have the lowest sleep requirements.*

Prediction 5. *The animals with the longest lifespans will have the lowest ROS accumulation rates, and these will be the same animals with the lowest sleep requirements.*

4 Evidence: The Full Comparative Record

4.1 Overview

Table 1 presents every known case of minimal or absent sleep mapped to basin geometry, lineage age, and maximum lifespan. The entries are ordered from deepest Lock to shallowest Explore.

Table 1: Complete Roster: Minimal/Absent Sleep
Mapped to Basin Geometry

Species	Lineage Age	Max. Lifespan	Sleep	Basin Type	Key Mechanism
<i>Nautilus</i> spp.	~500 Mya	Decades	None detected	Deepest Lock	Cold deep-water; zero social/nav complexity
<i>Limulus polyphemus</i> (horseshoe crab)	~450 Mya	20+ years	None detected	Deep Lock	Low metabolism; robust antioxidants; 5 mass extinctions survived
<i>Latimeria</i> spp. (coelacanth)	~400 Mya	Decades	Deep rest only	Deep Lock	600–700m depth; low metabolism; cave-dwelling
<i>Somniosus microcephalus</i> (Greenland shark)	~150 Mya	400–500 years	None observed	Deepest vertebrate Lock	Near-freezing metabolism; TMAO; minimal navigational complexity
<i>Rana catesbeiana</i> (bullfrog)	~200 Mya	16 years	Not detected; no rebound	Intermediate Lock	Ectotherm; low/variable metabolic rate; sit-and-wait predator

Continued...

Species	Lineage	Max. Lifespan	Sleep	Basin Type	Key Mechanism
<i>Sphenodon punctatus</i> (tuatara)	~200 Mya	100+ years	Two-state sleep; minimal	Moderate Lock	Low body temp (<20°C optimum); slow metabolism
<i>Proteus anguinus</i> (olm)	~50 Mya	100 years	Atypical; no circadian rhythm	Deep (cave) Lock	Perpetual dark; near-zero metabolism; no circadian N field
<i>Astyanax mexicanus</i> , cave populations	Same as surface (<10 Mya colonisation)	~5 years	Lost convergently ≥ 3 times	Experimentally confirmed deep Lock	Strongest proof: same genome as surface sleeper; sleep lost every time cave colonised
<i>Carcharodon carcharias</i> (great white shark)	~450 Mya (shark clade)	70+ years	No consolidated sleep possible	Constrained basin	Obligate ram ventilation; substitute rolling discharge
<i>Drosophila melanogaster</i> , min-sleep individuals	<0.1 Mya (population variant)	~60 days	Near-zero in some individuals; no fitness cost	Low-cost individual position	Within-species distribution confirms cost determines requirement

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Species	Lineage	Max. Lifespan	Sleep	Basin Type	Key Mechanism
<i>Ephydatia muel-leri</i> (sponge)	~600 Mya	Decades	Quiescence + re-bound	Cellular floor	Zero neurons; pure cellular maintenance; confirms pre-neural sleep origin
<i>Cassiopea</i> spp. (jellyfish)	~600 Mya	1–2 years	Sleep-like quiescence + rebound	Near-floor	Diffuse nerve net only; minimal synaptic component
<i>Hydra vulgaris</i>	~600 Mya	Potentially immortal	Quiescence + re-bound; light-regulated	Near-floor	Nerve net; light cue regulation
<i>Loxodonta africana</i> , wild matriarchs	Recent	65+ years	~2 hours/night, no re-bound	Extreme Explored (social max)	Sleep compressed to synaptic floor by maximal social navigational load
<i>Tachymarptis melba</i> (alpine swift)	Recent	~20 years	Minutes/day during 200-day flight	High-cost; rolling substitute	Unihemispheric rolling discharge; not flat cost
<i>Fregata minor</i> (great frigatebird)	Recent	~40 years	42 min/day in flight; 12h on land	High-cost; deferred discharge	Land rebound confirms cost is real; flight sleep is deferral not elimination

Continued...

Species	Lineage Age	Max. Lifespan	Sleep	Basin Type	Key Mechanism
<i>Zonotrichia leucophrys</i> (white-crowned sparrow), migratory	Recent	~13 years	70% reduction during migration; no cognitive impairment	Temporary basin transition	Migratory state = transient landscape shift; orexin elevated
<i>Homo sapiens</i>	~0.3 Mya	~120 years	7–9 hours REM-rich	Shallowest Explored	Maximum social/navigational/computational cost of any known species

4.2 The Decisive Proof: Cavefish Convergence

The *Astyanax mexicanus* cavefish data (North et al., 2024, bioRxiv/iScience) is the experiment that converts the theorem from strongly supported to effectively proven in the evolutionary domain [4].

The logic is unambiguous:

1. Surface *A. mexicanus* sleeps normally.
2. At least three cave systems were independently colonised by surface fish.
3. All cave populations show drastically reduced sleep with no rebound.
4. Sleep loss appears in hybrid populations, preceding morphological cave adaptation.
5. No fitness cost or aging penalty detected.
6. The wakefulness signal (orexin/hypocretin) is constitutively elevated — the same signal that transiently rises in migratory birds during their temporary basin transition.

The cave basin removes all three cost components simultaneously:

- No circadian N field (perpetual dark) $\Rightarrow C_1 \approx 0$
- Stable cold temperature; scarce, predictable food $\Rightarrow C_2 \approx 0$
- No predators; no social hierarchy; no spatial novelty $\Rightarrow C_3 \approx 0$

$$(C_1 + C_2 + C_3) \approx 0 \Rightarrow S_{\text{total}} \approx 0.$$

The genome did not change. The landscape changed. Sleep requirement tracks the landscape, not the genome.

4.3 The Greenland Shark: The Vertebrate Ceiling

Somniosus microcephalus [3] extends the theorem to its vertebrate maximum:

- Habitat: Arctic/sub-Arctic, -1°C to 4°C , depths to 2,200m.
- Lifespan: 272–512 years (radiocarbon dating of eye lens nucleus; median estimate ~ 392 years).
- Sexual maturity: ~ 150 years.
- Average swimming speed: 0.76 km/h.
- Sleep: No consolidated sleep observed. No published sleep architecture studies possible given habitat.
- Metabolism: Among the lowest of any vertebrate. Heart rate near the theoretical floor for a vertebrate organism.

The mechanism is thermodynamic. Near-freezing water temperature slows every chemical reaction by the Arrhenius equation — including oxidative damage cascades. TMAO provides protein stabilisation against pressure-induced unfolding and indirect antioxidant protection. The ROS production rate is matched or exceeded by passive clearance capacity without a dedicated discharge window.

The Greenland shark is not an exception to the theorem. It is the vertebrate proof of its limiting case:

$$\lim_{B \rightarrow \infty} \left[S(O), \frac{1}{L(O)} \right] = [0, 0] \quad (6)$$

No sleep. Maximum lifespan. One geometry.

4.4 The Frigatebird as the Control

The great frigatebird (*Fregata minor*) is the critical control that distinguishes genuine near-zero cost from deferred cost [7].

During oceanic flight: ~ 42 minutes sleep per day, primarily unihemispheric.

On land at the breeding colony: ~ 12 hours per day.

The elevated land sleep is a rebound discharge of accumulated flight-period cost. This confirms that the frigatebird's flight sleep is not cost elimination — it is deferral.

Contrast with the horseshoe crab: No rebound has ever been observed. No deprivation study has been possible because there is no measurable sleep state to deprive. The horseshoe crab does not defer cost. It does not accumulate cost. There is nothing to discharge.

The frigatebird is the control that makes the horseshoe crab result interpretable: the horseshoe crab is not suppressing sleep. It has nothing to suppress.

5 Evolutionary Convergences: The N Field Across Deep Time

The theorem makes a geometric prediction about evolutionary convergence: *independent lineages placed in the same landscape geometry will find the same solution, regardless of phylogeny.*

This prediction is confirmed at multiple biological levels:

5.1 Sleep Architecture Convergences

- **REM sleep:** Evolved independently in mammals and birds from a non-REM reptilian ancestor (~300 Mya divergence). Convergent because both crossed the cost threshold requiring offline memory consolidation independently.
- **Unihemispheric sleep:** Evolved independently in cetaceans, pinnipeds, and migratory birds. Convergent because all face the same constraint: cannot interrupt continuous locomotion or vigilance AND must discharge cost.
- **Sleep loss:** Evolved independently in at least 3 cavefish populations, the olm, and deep-water stable-niche invertebrates. Convergent because all entered basins that zeroed all three cost axes.

5.2 Circadian Clock Convergences

The circadian oscillator has been independently invented at least four times with completely different molecular components:

No shared protein homology across the four systems. The molecular hardware is arbitrary. The functional requirement — anticipating a 24-hour light-dark cycle imposed by the planet's rotation — is not.

Table 2: Independent Inventions of the Circadian Clock

Lineage	Components	Estimated Age
Cyanobacteria	KaiA/KaiB/KaiC	~2.7 Gya
Fungi (<i>Neurospora</i>)	FRQ/WC-1/WC-2	~1 Gya
Plants	CCA1/LHY/TOC1	~500 Mya
Animals	BMAL1/CLOCK/PER/CRY	~600 Mya

The N field here is Earth’s rotation itself. Every photosynthetic or aerobically metabolising organism on a rotating planet with a 24-hour light-dark cycle is subject to this N field. The attractor solution (anticipatory oscillator) is found four times independently because it is the only stable solution the landscape offers.

5.3 The Lock Strategy Across Geological Time

The deep-time record confirms the theorem’s most striking prediction: organisms that minimise their landscape position cost survive geological perturbations that eliminate the majority of other species.

Table 3: Lock Strategy Survival Across Mass Extinctions

Species	Lineage Age	Mass Extinctions Survived	Sleep
<i>Nautilus</i>	~500 Mya	5	None detected
<i>Limulus</i>	~450 Mya	5	None detected
<i>Latimeria</i>	~400 Mya	4	Deep rest only
Crocodylians	~200 Mya	2	Minimal/unihemispheric
Tuatara	~200 Mya	2	Minimal
<i>Homo sapiens</i>	~0.3 Mya	0	7–9 hours

The Permian extinction eliminated ~96% of all marine species. The horseshoe crab and nautilus survived unchanged. Their cost landscapes did not move enough during the event to force basin transition.

The Cretaceous-Paleogene extinction eliminated all non-avian dinosaurs. Crocodylians survived unchanged. Tuatara survived unchanged.

The Lock strategy is not just energetically efficient. It is **geometrically robust to perturbation**. An organism in a deep basin requires a very large perturbation to displace it. The mass extinctions that displaced most species were not large enough to displace these organisms’ niches.

6 The Unified Three-Variable Relationship

The theorem identifies three variables as co-expressions of one underlying geometric quantity: position-dependent cost accumulation rate.

$$\boxed{\mathcal{B}(O) \uparrow \iff S(O) \downarrow \iff L(O) \uparrow} \quad (7)$$

where $\mathcal{B}$ is basin depth, S is sleep requirement, and L is maximum lifespan.

These are not three independent correlations. They are the same correlation stated three different ways:

- $\mathcal{B} \uparrow \Rightarrow S \downarrow$: Deep basin = low cost accumulation = no discharge window needed.
- $\mathcal{B} \uparrow \Rightarrow L \uparrow$: Deep basin = low ROS accumulation = slow basal ageing rate = long lifespan.
- $S \downarrow \Rightarrow L \uparrow$: Low sleep requirement is a proxy for low ROS production, which is the same variable that determines lifespan.

The Greenland shark is the empirical proof of the limiting case: maximum basin depth, minimum sleep requirement (zero observed), maximum lifespan (400–500 years, the vertebrate record).

The human is the empirical proof of the other limit: minimum basin depth among long-lived organisms, maximum sleep requirement, lifespan ceiling at ~ 120 years despite the most sophisticated repair biology of any known organism.

The human is not sleeping too much. The human is paying the cost of being human.

7 Connection to the Identity Axis Theorem

The Identity Axis Theorem (IAT; Lawson 2026a) demonstrated that cancer is a cell displaced from its normal attractor basin into a false attractor, and that the maintenance mechanism of the false attractor (H, the convergence hub) is the drug target.

The Basin Depth–Sleep–Longevity Theorem operates at the organism level using the same geometric language.

The connection is not metaphorical. It is structural:

- In cancer: BMAL1/CLOCK is the molecular N field governing repair basin entry.

BMAL1 loss collapses the N field, makes the repair basin inaccessible, and locks the cell in the cancer false attractor.

- In sleep: BMAL1/CLOCK is the molecular N field governing repair basin entry at the cellular level. Chronic sleep deprivation degrades the G (epigenetic state), progressively eroding the depth of the repair basin until it becomes inaccessible.
- In longevity: The same repair basin that sleep maintains is the basin whose depth determines lifespan. The Greenland shark’s near-zero ROS accumulation means its cellular repair basin is never challenged. Its G does not degrade. It lives 400 years.

The prediction that follows — novel at time of writing:

Novel prediction from convergence of IAT and sleep geometry: IAT drug efficacy (EZH2 inhibitors, KDM1A inhibitors) will be inversely correlated with G degradation at identity anchor (A) gene loci. Patients with epigenomic evidence of repair basin erosion (BMAL1-low, G-degraded) will show reduced response to IAT-derived drugs. This is testable in existing trial datasets with matched epigenomic data.

8 What This Paper Is and Is Not

8.1 What this paper is

A deductive geometric framework for explaining:

1. Why sleep requirement varies by orders of magnitude across the animal kingdom.
2. Why the oldest lineages sleep the least.
3. Why the longest-lived vertebrate (Greenland shark) does not sleep.
4. Why sleep was lost convergently in at least three independent cavefish populations from the same genome.
5. Why lifespan, sleep requirement, and basin depth are co-expressions of the same underlying thermodynamic variable.

8.2 What this paper is not

- A statistical model. The framework generates hypotheses that are then tested statistically.
- A mechanistic neuroscience paper. The full molecular mechanisms of sleep are not derived here; the geometry that constrains them is.

- A description of all sleep biology. The framework describes the cost-discharge axis; sleep biology involves many additional dimensions.

8.3 Validation status at time of submission

- Predictions 1–5: All confirmed in the comparative record (Section 4).
- Falsification criteria: None triggered in the full dataset.
- Novel predictions generated: 1 (IAT drug efficacy + G degradation, Section 7).
- Geometric incompatibilities with prior framework documents: 0.

9 Summary and Conclusions

Sleep is a cost-discharge mechanism. Basin depth is the geometric variable that determines how much cost accumulates during waking. Lifespan is the biological readout of how slowly basal damage accumulates over a lifetime.

All three are expressions of the same quantity: an organism’s position in its attractor landscape.

The deepest-basin organisms — nautilus (500 Mya), horseshoe crab (450 Mya), Greenland shark (400-year lifespan) — do not sleep because they never generate the problem that sleep solves. Their cost landscapes are flat. Their ROS accumulation rates are matched by passive clearance. Nothing accumulates. Nothing needs discharging.

The shallowest-basin organism — the human — pays the full cost of its navigational complexity in sleep currency. Eight hours per night. Seven of every twenty-four hours discharged. The price of social cognition, language, memory, and the capacity to be displaced by every N field in the environment.

The cavefish (*Astyanax mexicanus*) is the proof. The same genome. Three independent cave colonisations. Sleep lost every time. Because the cave is a deep basin. And deep basins do not need sleep.

The theorem is not proven by these observations. It is supported by them to the point where the burden of proof has shifted to any alternative explanation that must account for all five predictions simultaneously without the geometry.

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Data Availability

All reasoning artifacts, prior framework documents, and derivation records are available at: <https://github.com/Eric-Robert-Lawson/attractor-oncology>

Prior Zenodo records in this series are cited below.

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key_claims:

- Sleep requirement = $f(\text{basin depth})$
- Lifespan = $f(\text{ROS floor}) = f(\text{basin depth})$
- These are the same equation at different timescales
- The Greenland shark is the vertebrate proof
 of the zero-sleep limiting case (400-year lifespan)
- The cavefish is the experimental proof
 (sleep lost ≥ 3 times independently, same genome)

connection_to_prior_work:

- IAT (cancer): BMAL1 as molecular N field for cellular repair basin entry
- Sleep topology T1-T8: topology classes are positions on the same cost spectrum formalised here
- Domestication: N field convergence at species level is the same geometric phenomenon at organism level

novel_predictions_generated: 1

- IAT drug efficacy inversely correlated with G degradation at identity anchor loci (testable in existing trial datasets)

repository:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>